Symmetries in Physics — Exercise Sheet 9

Representations, lifting, and conservation laws

Variant: Questions only

Exercise 1 Derive the Baker–Campbell–Hausdorff series to third order: expand $\log(e^{tX}e^{tY})$ in powers of t and collect the t^3 coefficient into commutators.

Exercise 2 Classify all continuous homomorphisms (i) $\varphi\colon \mathbb{R} \to \mathrm{U}(1)$ and (ii) $\Phi\colon \mathrm{U}(1) \to \mathrm{U}(1)$.

Exercise 3 Let $\gamma\colon [0,1] \to \mathrm{U}(1)$ be a continuous loop. Complete the branch-lifting construction of the winding number $w(\gamma)$ and prove it is a homotopy invariant.

Exercise 4 Compute the adjoint representation of the Heisenberg group H_3 on its Lie algebra, in the basis X, Y, Z with $[X, Y] = Z$ central. What is its kernel, and what does the answer compute ahead for Lecture 13?

Exercise 5 A free particle has $H = p^2/2m$. Show that the Galilean boost charge $G = tp - mx$ is conserved despite its explicit time dependence, and interpret the conservation law.

Exercise 6 Let Π be a unitary representation of a matrix Lie group on a finite-dimensional inner-product space, π its derivative. Show $\pi(X)^\dagger = -\pi(X)$ for all X, and that $A := \mathrm{i}\pi(X)$ is Hermitian — the physicists' observable.